Xi Li

Machine Learning Researcher — Multimodal AI for Spatial Biology and Computational Pathology

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Profile

Ph.D. researcher building multimodal machine-learning systems for spatial biology, computational pathology, and AI-assisted drug discovery. First author of **sMMC-22M** (NeurIPS 2026, accepted poster) and **STMoE** (CIKM 2026, accepted), and co-author of **TargetSage** (NeurIPS 2026, accepted poster). Experience scaling biomedical datasets beyond 20 million cell-aligned records, developing interpretable LLM reasoning for therapeutic target discovery and single-cell gene-expression prediction, and translating research prototypes into deployed products.

Technical Strengths

Machine Learning	Multimodal representation learning, mixture-of-experts, transformers, graph neural networks, diffusion models, contrastive learning, retrieval, domain generalization
Biomedical AI	Computational pathology, H&E/whole-slide imaging, spatial transcriptomics, single-cell genomics, cell-level prediction, biomedical foundation models
Engineering	Python, PyTorch, multi-GPU experimentation, large-scale dataset curation, reproducible benchmarking, scientific visualization

Education

University of California, Irvine — **Irvine, CA**
Ph.D. in Computer Science — *Sep 2023 – Present*

Tsinghua University — **Beijing, China**
M.S. in Computer Science and Technology — *Sep 2019 – Jun 2022*

University of Science and Technology of China — **Hefei, China**
B.S. in Information Security — *Sep 2015 – Jun 2019*

Selected Research Experience

University of California, Irvine — **Irvine, CA**
Ph.D. Student Researcher — *Sep 2023 – Present*

- Led **STMoE**, a multi-scale mixture-of-experts framework for WSI-only, full-gene, single-cell expression prediction; paired local and tissue-context crops with frozen UNI2-h representations and organ-routed retrieval. Accepted at **CIKM 2026**.
- Curated **CellST-24M**, aligning 23.94 million cells across 66 studies and 25 organ categories, and built leakage-controlled in-domain, cross-patient, cross-platform, and multi-organ benchmarks.
- Built **sMMC-22M**, a context-aware multimodal resource whose current manifest contains more than 28 million aligned targets across 25 tissue categories; designed cell-level benchmarks for scale, biological context, and transfer robustness. Accepted as a poster at **NeurIPS 2026**.
- Contributed to **TargetSage**, an **AI-assisted drug discovery** framework for therapeutic target identification, combining agentic biological-evidence profiling (PHAROS, GWAS, CTDbase), interpretable LLM reasoning, and positive–unlabeled learning to prioritize candidate target genes under incomplete labels. Accepted as a poster at **NeurIPS 2026**.
- Developed **iHAST**, an arbitrary-scale hybrid-attention model for spatial-transcriptomics super-resolution, outperforming prior methods across more than 800 multi-platform examples (BMVC 2024).
- Developed reliability and spatial-perturbation modeling frameworks for biomedical foundation models; two first-author manuscripts are under review at **KDD 2027**.

Multimedia Lab, Tsinghua University — **Beijing, China**
Researcher, AI Broadcast Production — *Oct 2021 – Sep 2022*

- Built a cross-attention system that fused audio and video signals to automate multi-camera switching, crop selection, and virtual camera motion for live orchestral performances.
- Deployed the system at China's National Centre for the Performing Arts, translating a multimodal research prototype into a real production workflow.

Industry and Product Leadership

ByteDance AI Lab **Beijing, China**
AI Product Manager Intern *May 2021 – Dec 2021*
- Led product planning for an internal generative-imaging platform; coordinated research, engineering, and design teams to move image-generation and editing models into scalable creative tools.

ByteDance — Feishu/Lark **Beijing, China**
Data Product Intern *Jun 2020 – Nov 2020*
- Improved enterprise document and knowledge retrieval through behavior analytics, relevance analysis, and data-informed search-product design.

Hantao Zhiying **China**
Chief Product Lead *2022 – 2023*
- Owned product strategy and delivery for AI imaging products, from feature definition and user experience through technical coordination and deployment.

Nolibox **Beijing, China**
Partner and Co-founder *Mar 2020 – Aug 2020*
- Developed an automatic logo-generation system and font-stroke generation tools; the company received a Red Dot Award and raised multimillion-dollar investment.

Mentoring and Open-Source Leadership

Google Summer of Code (GSoC) **2025 & 2026**
Open-Source Research Mentor
- **2026 — Omni-ST: Instruction-Driven Any-to-Any Multimodal Modeling for Spatial Transcriptomics**. Mentored project design across histology, gene expression, spatial graphs, and text for cross-modal prediction, cell typing, and biological interpretation (350-hour project scope).
- **2025 — Disentangled Generation and Editing of Pathology Images**. Mentored generative modeling and latent-space editing to control tumor grade, malignant-cell counts, and staining while preserving realistic cell distributions (350-hour project scope).

Selected Publications

1. **Xi Li**, Yaqi Hu, Ziheng Duan, Chaoyang Wang, and Jing Zhang. "STMoE: Multi-Scale Mixture-of-Experts for Single-Cell Gene Expression Prediction from Histology." **CIKM 2026, accepted**. doi:10.1145/3799682.3841080.
2. **Xi Li**, Yaqi Hu, Ziheng Duan, Diptanshu Sikdar, Yang Liu, and Jing Zhang. "sMMC-22M: A Context-Aware Dataset and Benchmark for Single-Cell Spatial Transcriptomics." **NeurIPS 2026, accepted as a poster**.
3. Ziheng Duan, Tong Wu, **Xi Li**, Simon Dongbo Sun, Chaoyang Wang, Junhao Liu, Martin Renqiang Min, and Jing Zhang. "TargetSage: Identifying Therapeutic Target Genes with Interpretable and Robust LLM Reasoning." **NeurIPS 2026, accepted as a poster**. OpenReview.
4. Liyun Lin, Shuozhen Bao, **Xi Li**, Isaiah Yim, Dejiang Wang, Gaocai Li, Xiangjun Di, Shujie He, Junyi Xu, Kathryn Miller-Jensen, Jing Zhang, and Rong Fan. "Spatially Indexed Single-Cell Morphology and Transcriptome Profiling by DBiC-seq." **Nature Communications**, *under review.*
5. **Xi Li**, Ziheng Duan, Yi Dai, Yang Liu, and Jing Zhang. "Context Is Not a Side Note: A Support-Calibrated Reliability Framework for Biomedical Foundation Models." **KDD 2027, under review**.
6. **Xi Li**, Ziheng Duan, Simon Dongbo Sun, Martin Renqiang Min, Yang Liu, and Jing Zhang. "Spine: Disentangling Tissue Context and Perturbation-Conditioned Response Fields in Spatial Genomics." **KDD 2027, under review**. OpenReview.
7. "CCR2 Deficiency Attenuates Flow-Induced Carotid Artery Remodeling." **Vascular Pharmacology**, *under review.* Co-authors include Fu Gao and **Xi Li**; corresponding author: Yang Liu.
8. Fu Gao, Ian Mason, Mingze Dong, Yao Lu, Di Zhang, Xing Lou, **Xi Li**, Irbaz Hameed, Mingyu Yang, Mei Zhong, Markus Krane, Giovanni Ferrari, George Tellides, Yang Liu, Rong Fan, and Arnar Geirsson. "Cross-Species Multi-Omics Identifies a Conserved Fibrogenic Valvular Interstitial Cell Population across Mitral Valve Diseases." **Circulation**, *under review; editorial consideration.*
9. **Xi Li**, Ziheng Duan, Yi Dai, Siwei Xu, and Jing Zhang. "iHAST: Integrating Hybrid Attention for Super-Resolution in Spatial Transcriptomics." **BMVC 2024**.
10. Ziheng Duan*, **Xi Li***, Zhiqing Xiao, Rex Ying, and Jing Zhang. "MUSE: A Multi-Slice Joint Analysis Method for Spatial Transcriptomics Experiments." **CIKM 2025**. *Equal contribution.
11. Ziheng Duan, **Xi Li**, Zhuoyang Zhang, James Song, and Jing Zhang. "DISCO: A Diffusion Model for Spatial Transcriptomics Data Completion." **ICIP 2025**.